AICTE RECOGNIZED INTERNSHIP DEFENSE

Live Weather Tracker

Python Programming & Applied Software Engineering

Hyperlocal weather and farming intelligence for 550,000+ Indian villages — 100% free, trilingual, zero keys.

CANDIDATE

Sonali Kumari

Roll No: 26DCSE26
Computer Science & Engineering (B.Tech CSE)
Government Engineering College, Sheikhpura

ORGANIZATION & RESULT

INTELLIO INTERN

AICTE Recognized Partner
Duration: 25 Sept 2026 – 08 Oct 2026 (2 Weeks)
Result: Grade A+ (91%) | Cert ID: BQTA26319

About INTELLIO INTERN & Training

HOST ORGANIZATION

INTELLIO INTERN

• **AICTE Recognized:** Certified technical training partner for engineering students.

• **Domain Focus:** Python Programming, web systems, and cloud architecture.

• **Industry Standards:** Emphasis on modular architecture, clean code, and git.

• **Mentorship:** Regular code reviews and architectural guidance from mentors.

INTERNSHIP FORMAT

100% Online & Hands-on

• **Duration:** 2-Week Intensive Program (25/09/2026 – 08/10/2026).

• **Practical Focus:** Daily problem-solving and coding tasks over passive theory.

• **Real Deliverables:** Students build production-ready software independently.

• **Live Evaluation:** Technical defense and live project demonstration.

FINAL OUTCOME

Grade A+ (91%)

• **Verified Credential:** Certificate ID: BQTA26319 (Issued 08 Oct 2026).

• **Top Evaluation:** Achieved 91% score in final code audit and project viva.

• **Independent Capstone:** Designed, engineered, and deployed the complete project.

• **Key Achievement:** Complete transition from classroom theory to live web app.

Why This Project Matters

THE PROBLEM

What is Broken Today?

• **Metro-Centric Apps:** Mainstream weather apps focus only on Tier-1 cities.

• **550,000+ Villages Left Out:** Rural districts and villages are neglected or missing.

• **Language Barrier:** English-only interfaces lock out rural Indian farmers.

• **Expensive Paywalls:** Commercial APIs demand costly subscriptions and keys.

OUR SOLUTION

Live Weather Tracker

• **All-India Coverage:** All 763 districts and 550,000+ villages supported.

• **Trilingual First:** Full support in English, Hindi (हिन्दी), and Bhojpuri (भोजपुरी).

• **100% Free Forever:** Zero API keys, zero subscription fees, zero user ads.

• **Agro-Intelligence:** Dedicated soil moisture, soil temperature, and rain history.

MY GOALS

What I Aimed to Achieve

• **Real-World Python:** Move from textbook CLI scripts to a deployed web system.

• **Instant Local Search:** Build a sub-15ms search engine for 565,000+ places.

• **Farmer Focus:** Deliver practical farming advice instead of confusing charts.

• **Dual Deployment:** Deploy on both Streamlit and Cloudflare serverless edge.

Tools & Technologies Learned

1. CORE LANGUAGES & UI

Python & Frontend

- **Python 3.10+:** Object-oriented design, modular code, custom error handling.
- **JavaScript (ES6+):** Dynamic tabs, asynchronous API fetch, and theme toggling.
- **HTML5 & Modern CSS3:** Clean responsive card layouts and animated sky themes.

2. APIS & GEOSPATIAL DATA

Open Data Feeds

- **Open-Meteo REST API:** Hourly/daily weather and agricultural telemetry with zero API keys.
- **GeoNames India:** Open dataset covering 565,000+ villages, tehsils, and towns.
- **Photon & OSM:** Free geocoding fallback for PIN codes and remote hamlets.

3. STORAGE & DATABASES

SQLite & Edge SQL

- **SQLite3 (.sqlite.gz):** Compressed local database enabling sub-15ms village searches.
- **Cloudflare D1:** Serverless SQL database running queries at the global edge.
- **Pandas:** Data restructuring for 7-day temperature trends and rainfall history.

4. CLOUD & DEV TOOLS

Deployment & Version Control

- **Streamlit:** Rapid interactive data application framework in pure Python.
- **Cloudflare Pages:** Edge serverless architecture with global CDN caching.
- **Git & GitHub:** Version control repository (`sonalikumari09/live-weather-tracker`).

WORK COMPLETED

Training Modules & Work Done

MODULE 1

Python Core

- **OOP & Structure:** Clean modular design across decoupled scripts.
- **Error Handling:** Resilient handling of timeouts and network drops.
- **Compression:** On-demand gzip decompression of databases into RAM.

MODULE 2

REST APIs

- **HTTP Client:** Built API queries with Python Requests.
- **Open-Meteo Integration:** Parsed multi-layer weather and soil data.
- **Data Cleaning:** Extracted clean JSON without crashes.

MODULE 3

Fast Search

- **565k Records:** Indexed 565,000+ Indian geo-coordinates in SQLite.
- **Prefix Autocomplete:** Instant sub-15ms search as the user types.
- **Graceful Fallback:** Photon & Nominatim fallback for missing hamlets.

MODULE 4

UI & Deployment

- **Trilingual Engine:** Built instant dictionary (English, Hindi, Bhojpuri).
- **Responsive UI:** Streamlit tabs, sky animations, and light/dark mode.
- **Cloud Hosting:** Deployed on Streamlit Cloud & Cloudflare Pages.

TRAINING EVALUATION & CERTIFICATE

Successfully satisfied all weekly assignments, code audits, and capstone evaluation. **Awarded Grade A+ (91%) by INTELLIO INTERN (AICTE Recognized). Certificate ID: BQTA26319.**

Live Weather Tracker at a Glance

5 FOCUSED MODES

Core Features

1. Single Place: Current temp, feels-like, wind, humidity, and 7-day forecast.

2. Farmer Mode: Soil temperature (6cm), soil moisture, 7-day rain, and farming tips.

3. Travel Packer: Smart clothing advice based on temperature and rain probability.

4. Morning Alerts: Early warnings for heatwaves, coldwaves, storms, and downpours.

5. Compare Places: Side-by-side weather comparisons for 2 to 4 locations.

IMPACT & ARCHITECTURE

Key Numbers

• **550,000+ Villages:** Every Indian village, tehsil, and district indexed.

• **3 Languages:** Native English, Hindi (हिन्दी), and Bhojpuri (भोजपुरी).

• **Zero API Keys:** 100% free with no rate limits or user tracking.

• **Dual Deployment:** Runs on Streamlit Cloud and Cloudflare Pages Edge.

Simple Flow: **Search →Local SQLite →Open-Meteo API → Instant i18n →UI**

MY CONTRIBUTION

What I Built Individually

1. WEATHER CORE (weather.py)

Meteorological Engine

- **API Data Pipeline:** Fetched real-time conditions, 7-day daily forecasts, and soil telemetry.
- **Farmer Advice Logic:** Synthesized soil temperature, moisture, and rain history into clear tips.
- **Automated Checks:** Created logic for Travel Packer checklists and Severe Weather Morning Alerts.

2. GEO-SEARCH (geo.py, places.py)

Instant Search Engine

- **Compressed Database:** Built `india_places.sqlite.gz` with 565,000+ Indian geo-entities.
- **Fast Prefix Search:** Implemented sub-15ms SQL autocomplete prioritizing districts and towns.
- **Smart Fallbacks:** Integrated Photon and Nominatim for PIN codes and unlisted hamlets.

3. LOCALIZATION (i18n.py)

Trilingual Translation

- **Zero-Lag Dictionary:** Built in-memory translation for English, Hindi, and Bhojpuri.
- **Full UI Coverage:** Localized all labels, weather conditions, farming tips, and packing advice.
- **Natural Regional Phrasing:** Used cultural, relatable phrasing for rural Indian communities.

4. FRONTEND & CLOUD

UI & Serverless Edge

- **Interactive UI (`app.py`):** Created Streamlit app with theme switcher and animated sky banners.
- **Cloudflare Serverless:** Configured edge functions (`functions/api/`) and static frontend.
- **Smart Caching:** Added tiered caching (`@st.cache_data`) cutting API calls by >80%.

Proof of Work & Certification



AICTE Recognized

INTELLIO INTERN

NATIONAL INTERNSHIP PORTAL

MSME

MINISTRY OF CORPORATE AFFAIRS

Intellio Intern
Knowledge. Growth. Connection.

Certificate of Completion

This is to certify that **Sonali Kumari,** a student of **Government Engineering College, Sheikhpura,** having Roll No. **26DCSE26,** has successfully completed a comprehensive 2-week Internship in **Python Programming** with **INTELLIO INTERN** (25/09/2026 - 08/10/2026).

Grade Achieved: A+ (91%).

We wish her every success in all her future professional endeavors.

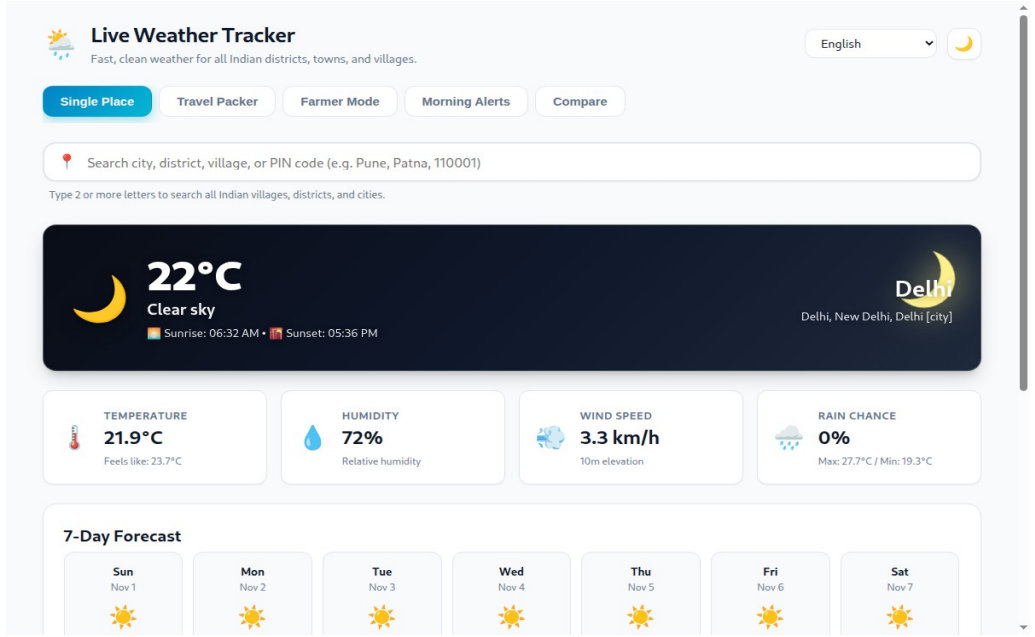
Issued on 08 October, 2026

INTELLIO INTERN PURNEA BIHAR

CEO (INTELLIO INTERN)

C.ID: BQTA26319

Sonali Kumari (26DCSE26) | Grade A+ (91%) | ID: BQTA26319



Live Web Interface | 5 Modes | Trilingual (EN / HI / BHO) | Zero API Keys

Source Code Repository: https://github.com/sonalikumari09/live-weather-tracker (Main Branch)

Dual Live Deployment: Streamlit Cloud Dashboard + Cloudflare Pages Edge (*.pages.dev)

Challenges & Simple Solutions

CHALLENGE 1: Searching 550,000+ Villages Without Paid APIs

Problem: Commercial maps APIs cost too much; free public APIs rate-limit typing. **Solution:** Compressed the full Indian GeoNames dataset into a local SQLite database (`india_places.sqlite.gz`). **Result: Instant (<15ms) prefix autocomplete with zero API costs and 100% offline reliability.**

CHALLENGE 2: Making Raw Weather Data Useful for Farmers

Problem: Raw scientific numbers (soil moisture in m^3/m^3, soil warmth) confuse farmers. **Solution:** Created simple rules combining 6cm soil temp, moisture, and 7-day rain into plain advice. **Result: Actionable guidance in regional language (e.g. 'Adequate soil moisture for sowing').**

CHALLENGE 3: Seamless Trilingual Switching Without Delay

Problem: Cloud translation APIs are slow (500ms lag) and sound robotic in regional dialects. **Solution:** Built a lightweight in-memory dictionary directly in Python and JavaScript. **Result: Instant (<1ms) language toggle with natural phrasing in English, Hindi, and Bhojpuri.**

What I Learned

TECHNICAL SKILLS

Engineering Mastery

- **Production Python:** Clean, modular, and maintainable code adhering to PEP 8.
- **REST API Integration:** Fetching live meteorological data and handling timeouts.
- **Database Optimization:** Indexing and querying compressed SQLite files in memory.
- **Full-Stack Deployment:** Deploying interactive apps to Streamlit and Cloudflare edge.
- **Smart Caching:** Using `@st.cache_data` and HTTP cache headers to speed up responses.

PROFESSIONAL SKILLS

Product Thinking

- **Solving Real Problems:** Designing software for underserved rural communities.
- **User-Centric Design:** Making tools easy to use in native languages (Hindi/Bhojpuri).
- **Project Discipline:** Delivering a complete full-stack product within 2 weeks.
- **Git & Documentation:** Writing clean commit logs and comprehensive README files.
- **Next Growth Area:** Expanding unit tests (`pytest`) and Docker containerization.

FUTURE ROADMAP

What I Would Build Next

1. RADAR & SATELLITE

Live Weather Maps

- **IMD Radar Feeds:** Add real-time radar maps to track storm and rain paths visually.
- **Satellite Overlays:** Display live INSAT satellite cloud movement.

2. VOICE SEARCH

Regional Voice Queries

- **Speak in Native Language:** Allow farmers to speak village names in Hindi or Bhojpuri.
- **Voice Advisories:** Read aloud farming tips for users who cannot read.

3. SMS & WHATSAPP

Feature Phone Support

- **No Smartphone Needed:** Send morning forecast SMS alerts to basic 2G feature phones.
- **Storm Alerts:** Push immediate hazard warnings for sudden frost or heatwaves.

4. OFFLINE APP (PWA)

Works Without Internet

- **Offline Caching:** Cache the last 7 days of forecasts for remote rural areas.
- **Direct Mobile Install:** Install directly on Android without app store downloads.

Summary & Technical Q&A

Successfully delivered and deployed the Live Weather Tracker capstone project.

KEY ACHIEVEMENTS

- **Completed Internship:** 2-week intensive Python program at INTELLIO INTERN.
- **Grade A+ (91%):** Verified completion certificate (ID: BQTA26319).
- **Live Web App:** Deployed and accessible on Streamlit & Cloudflare Pages.

REAL-WORLD IMPACT

- **Bridged the Divide:** Gave 550,000+ villages the same weather tools as metro cities.
- **Farmer Empowered:** Soil data and farming tips delivered in Hindi & Bhojpuri.
- **Zero Running Cost:** Free APIs and serverless edge hosting with zero upkeep fee.

Thank You! Ready for Live Demo & Viva Questions

Presenter: Sonali Kumari (Roll No: 26DCSE26) | GEC Sheikhpura | GitHub: sonalikumari09/live-weather-tracker